

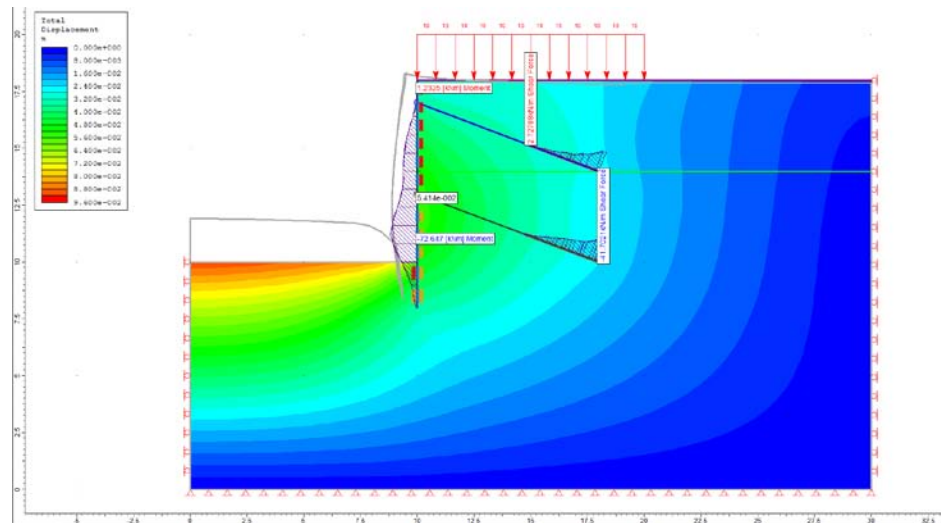
Anchored Sheet Pile Wall Tutorial

In this tutorial, *Phase2* is used to simulate the construction of an excavation supported by a sheet pile wall anchored with grouted tiebacks.

The complete model can be found in the **Tutorial 23 Anchored Sheet Pile Wall.fez** file. All tutorial files installed with *Phase2* 8.0 can be accessed by selecting File > Recent Folders > Tutorials Folder from the *Phase2* main menu.

Topics covered

- Liners
- Joints
- Structural interface
- Grouted tieback bolts
- Surcharge pressure



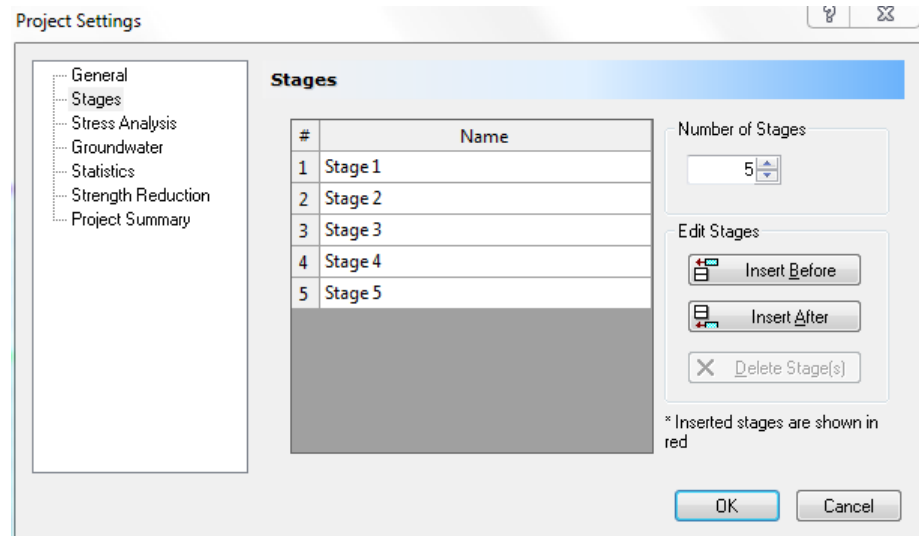
Model

Start the *Phase2* Model program.

Project Settings



Open the **Project Settings** dialog from the **Analysis** menu and make sure the General tab is selected. Define the units as being “Metric, stress as kPa”. Select the Stages tab and change the number of stages to 5. Close the dialog by clicking OK.



Geometry

The problem consists of two soil layers and two stages of excavation. Therefore an external boundary, a material boundary, and a stage boundary are required as shown below.



Start by creating a rectangular external boundary. Select the **Add External** option in the **Boundaries** menu and enter the following coordinates:

0 , 0

30 , 0

30 , 18

20 , 18

0 , 18

c (to close the boundary)



Now we need to delineate the different material layers within the model. Go to the **Boundaries** menu and select **Add Material**. Enter the following points:

0 , 14

30 , 14

Hit Enter to finish entering points.

Next we will define the boundaries of the two stages of excavation. We do not need to explicitly define the bottom of the first excavation since this is coincident with the material boundary. Therefore we will just draw the bottom of the second excavation and the vertical line that defines the lateral extent of the excavation.



From the **Boundaries** menu, select **Add Stage** (we use a stage boundary instead of an excavation boundary because an excavation boundary requires a fully enclosed space). Enter the following points:

0 , 10

10 , 10

10 , 18

Enter

The model should now look like this:



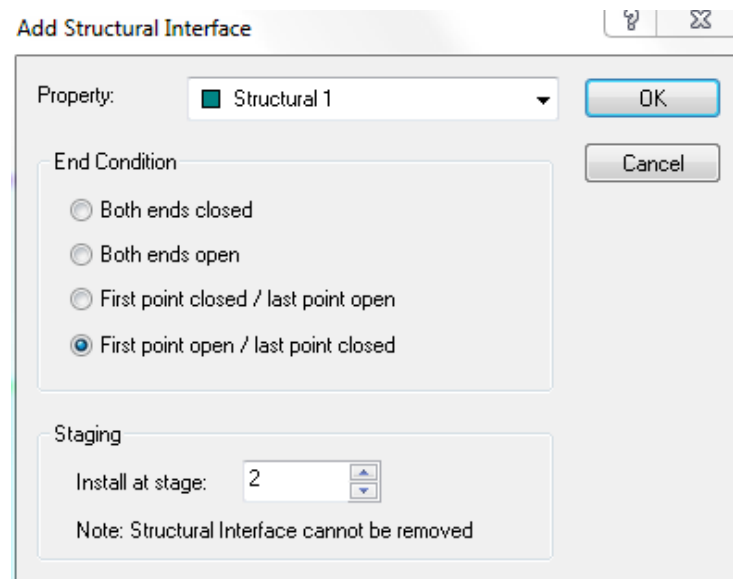
Sheet pile wall

Before we generate the mesh, we need to define the boundary that delineates the sheet pile wall. The top part of the wall will be coincident with the excavation but the wall will extend below the excavated soil.

We wish to allow slip between the soil and the wall. Therefore we will need a sliding interface (joint) on both sides of the wall. The way to do this in *Phase2* is to use a Structural Interface boundary.



The wall will be installed in Stage 2, so first click on the Stage 2 tab at the bottom of the screen. Go to the **Boundaries** menu and select **Add Structural Interface**. You will see the Add Structural Interface dialog. We want to set up the interface so that the top is open (the joint ends are free to slip past each other) and the bottom is closed (the joint ends are attached and cannot slip past each other). Assuming that we draw the interface from the top down, select the option for 'First point open / last point closed'. Ensure that 'Install at stage:' equals 2 as shown.



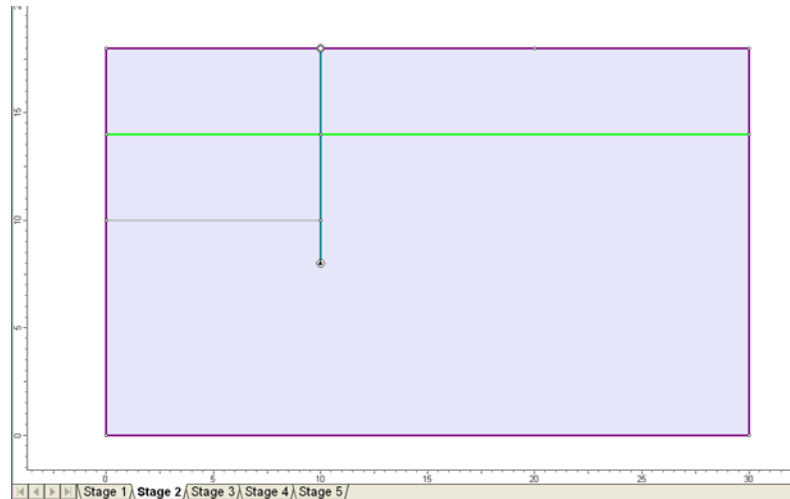
Click OK and you can enter points to define the structural interface. Enter the following points:

10 , 18 (or click on the existing point on the top boundary)

10 , 8

Enter

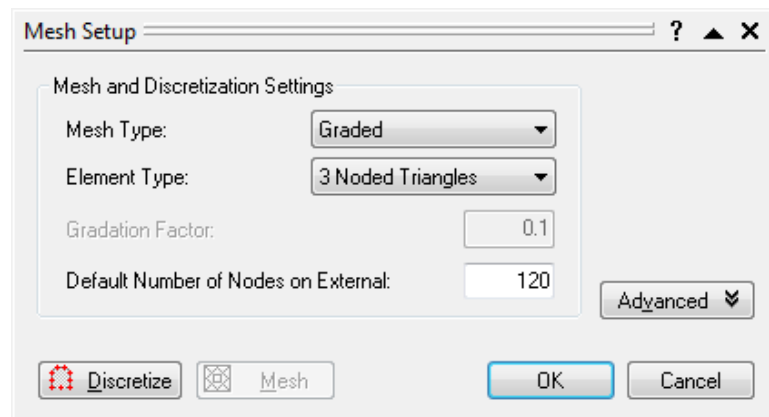
The model should now look like this:



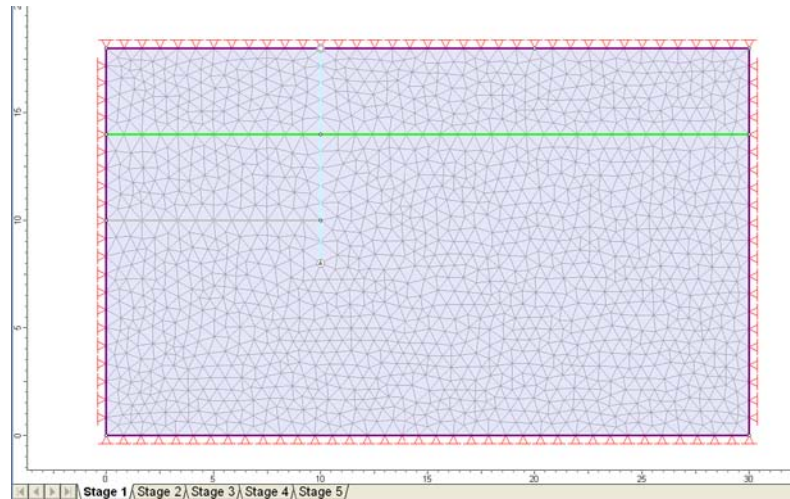
Mesh



Go back to Stage 1. Add the finite element mesh by selecting **Mesh Setup** from the **Mesh** menu. In the mesh setup dialog, change the Default Number of nodes to 120 as shown.



Click the Discretize button and then the Mesh button. Click OK to close the dialog. The mesh will look like this:



Boundary conditions



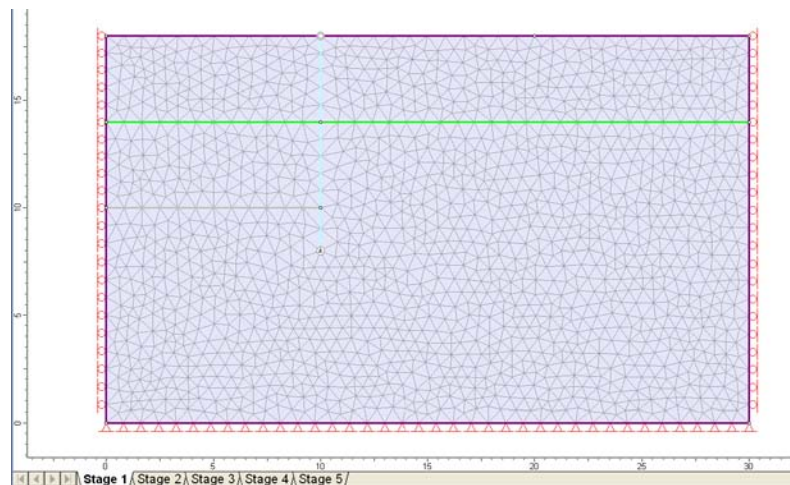
By default, all segments of the external boundary are fixed. Since the top of this model represents the actual ground surface, we need to free the top surface. Go to the **Displacements** menu and select **Free**. Click on the three sections that make up the top boundary and hit Enter. You will see that the fixed boundary conditions have disappeared from the top boundary.



The left and right edges should be fixed only in the x-direction to allow vertical movement. Select **Restrain X** from the **Displacements** menu and select all the sections of the left and right boundaries. These boundaries will now be showing rollers instead of pins.



Finally, we need to re-establish the fixed boundary condition on the bottom corners. Select **Restrain X,Y** from the **Displacement** menu, click on the bottom boundary and hit Enter. Your model should now look like this:



Field Stress



Because the top of the model represents the true ground surface, we want to use a gravity field stress. Go to the **Loading** menu and select **Field Stress**. For Field Stress Type select Gravity and click the check box for “Use actual ground surface”. Leave all other values as default.

Field Stress Properties

Field Stress Type: **Gravity**

☒ Use actual ground surface ☐ Use effective stress ratio ☐ Use variable stress ratio

Ground Surface Elevation (m):

Unit Weight of Overburden (kN/m³):

Total Stress Ratio (horiz/vert in plane):

Total Stress Ratio (horiz/vert out-of-plane):

Locked-in horizontal stress (in plane) (kPa, Comp. +):

Locked-in horizontal stress (out-of-plane) (kPa, Comp. +):

OK Cancel Statistics... Advanced >>

Click OK to close the dialog.

Materials



We now need to define the material properties and assign the correct materials to the correct parts of the model. Go to the **Properties** menu and select **Define Materials**. Change the name of Material 1 to Sand. Enter the other material parameters as shown.

Define Material Properties

Sand

Name: Sand Material Color:

Initial Element Loading: **Field Stress & Body Force** Unit Weight: (kN/m³):

Elastic Properties

Elastic Type: **Isotropic** Poisson's Ratio:

Young's Modulus (kPa): Young's Modulus (resid) (kPa):

E1 (kPa): E2 (kPa): E3 (kPa):

ν_{12} : ν_{13} : ν_{23} :

Strength Parameters

Failure Criterion: **Mohr Coulomb** Material Type: **Plastic**

Tensile Strength (peak) (kPa): Dilatation Angle (deg):

Fric. Angle (peak) (deg): Fric. Angle (resid) (deg):

Cohesion (peak) (kPa): Cohesion (resid) (kPa):

Tensile Strength (resid) (kPa):

☐ Stage Properties ☐ Datum Dependent

Define Factors... Define Properties...

Unsaturated Shear Strength

Phi b: Air Entry (kPa):

Copy To... Statistics... ☐ Show only properties used in model OK Cancel

Click on the tab for Material 2, change the name to Clay and enter the following properties:

The dialog box is titled "Define Material Properties". On the left is a list of materials from Sand to Material 29. The "Clay" material is selected. The main panel shows the following settings:

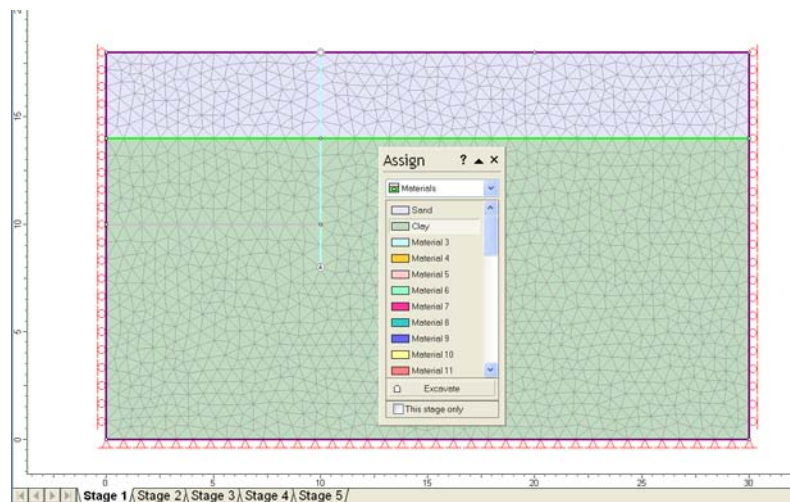
- Name:** Clay
- Material Color:** Green
- Initial Element Loading:** Field Stress & Body Force
- Unit Weight: (kN/m3):** 20
- Elastic Properties:**
 - Elastic Type:** Isotropic
 - Poisson's Ratio:** 0.25
 - Young's Modulus (kPa):** 15000
 - Young's Modulus (resid) (kPa):** 20000
 - E1 (kPa):** 20000, **E2 (kPa):** 20000, **Ez (kPa):** 20000
 - v12:** 0.2, **v1z:** 0.2, **v2z:** 0.2
- Strength Parameters:**
 - Failure Criterion:** Mohr Coulomb
 - Material Type:** Plastic
 - Tensile Strength (peak) (kPa):** 0
 - Dilation Angle (deg):** 0
 - Fric. Angle (peak) (deg):** 25
 - Fric. Angle (resid) (deg):** 25
 - Cohesion (peak) (kPa):** 10
 - Cohesion (resid) (kPa):** 10
 - Tensile Strength (resid) (kPa):** 0
- Stage Properties:** (unchecked)
- Datum Dependent:** (unchecked)
- Unsaturated Shear Strength:**
 - Phi b:** 0
 - Air Entry (kPa):** 0

Buttons at the bottom include "Copy To...", "Statistics...", "Show only properties used in model", "OK", and "Cancel".

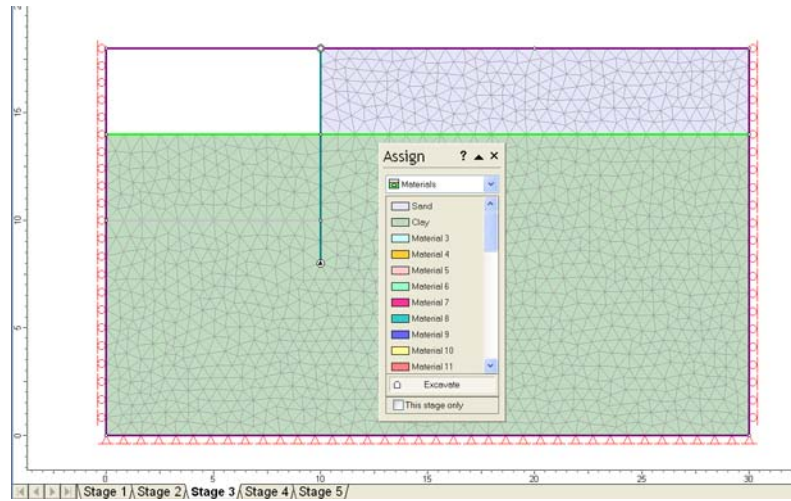
Click OK to close the dialog.



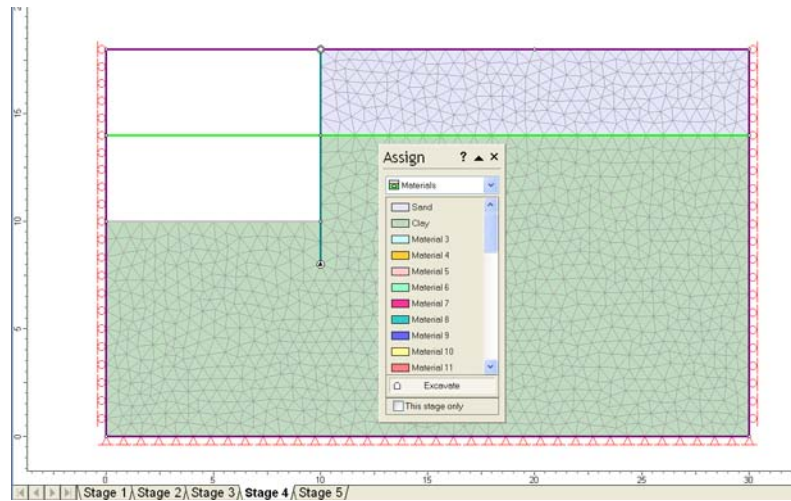
To assign the materials to the model, select **Properties → Assign Properties**. Be sure you are looking at the first stage. By default, everything is set to Sand material. Select Clay from the assign dialog and click in the sections of the model below the green material boundary.



This sets up the initial state for the first stage. In the second stage, we will add support so there is no change in material. In the third stage we will start excavating. Click on the Stage 3 tab. Choose Excavate from the Assign menu and click inside the top left section of the model as shown:



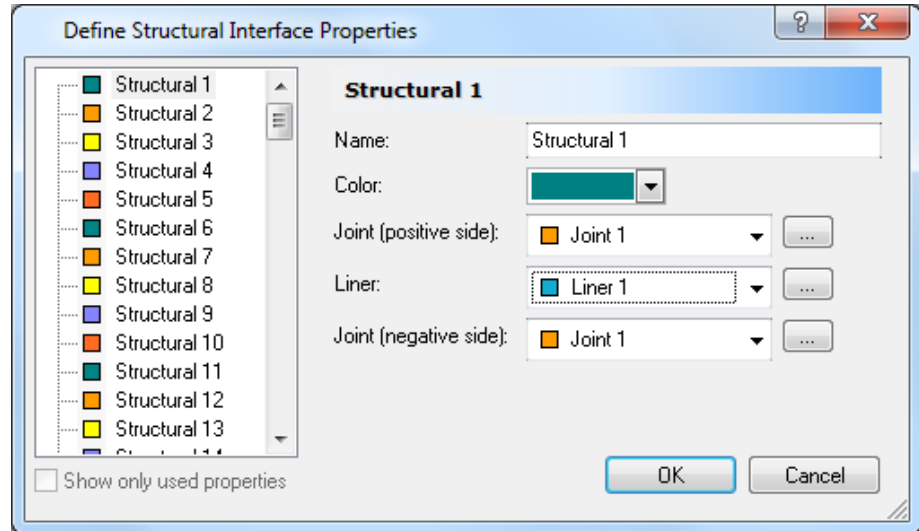
In Stage 4 we will install another tieback and excavate further. Click on the tab for Stage 4 and then click inside the next area to be excavated as shown.



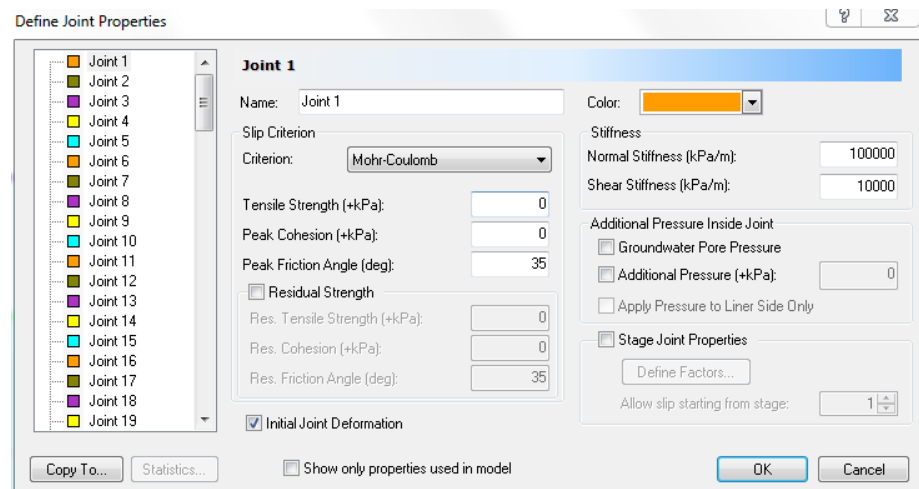
Close the Assign dialog and click through the stages to ensure that the excavation proceeds correctly.

Sheet pile wall

The sheet pile wall will be installed in Stage 2 so click on the tab to show Stage 2. In our model, the sheet pile wall is sandwiched between two joints. The wall plus the joints together make up a structural interface. Right click on the structural interface (dark green line) and select Structural Interface Properties. You will now see a dialog that gives the default properties of the liner (wall) and the two joints.

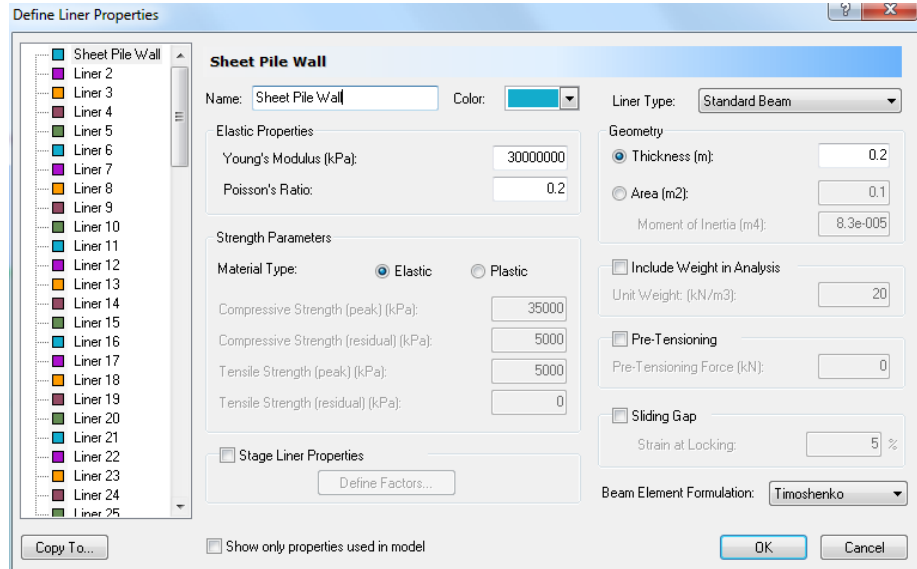


To modify the properties of the joints, click on the button (...) to the right of Joint 1. We want to allow slip on the joint so change the Slip Criterion to Mohr-Coulomb. Leave all other properties as the default properties as shown.



Click OK to close the dialog and go back to the Structural Interface dialog. Click on the button (...) next to Liner 1 to set its properties.

In the Define Liner Properties dialog, change the name to Sheet Pile Wall. Set the thickness to 0.2 m as shown.



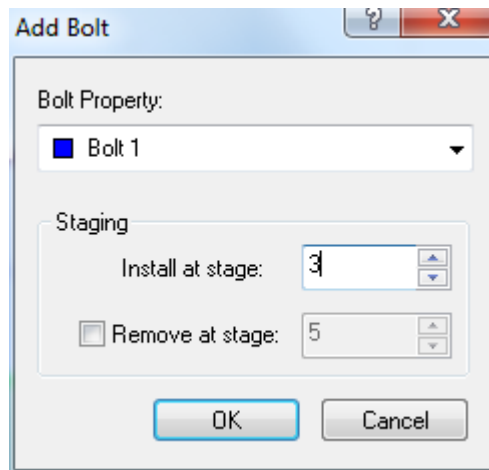
The 'Define Liner Properties' dialog box is shown. On the left is a list of 25 liners, with 'Sheet Pile Wall' selected. The main area is titled 'Sheet Pile Wall'. It contains several sections: 'Name' (Sheet Pile Wall), 'Color' (blue), 'Liner Type' (Standard Beam), 'Elastic Properties' (Young's Modulus: 30000000, Poisson's Ratio: 0.2), 'Strength Parameters' (Material Type: Elastic, Compressive Strength (peak): 35000, Compressive Strength (residual): 5000, Tensile Strength (peak): 5000, Tensile Strength (residual): 0), 'Geometry' (Thickness: 0.2, Area: 0.1, Moment of Inertia: 8.3e-005), 'Include Weight in Analysis' (Unit Weight: 20), 'Pre-Tensioning' (Pre-Tensioning Force: 0), 'Sliding Gap' (Strain at Locking: 5%), and 'Beam Element Formulation' (Timoshenko). There are 'Copy To...', 'Show only properties used in model', 'OK', and 'Cancel' buttons at the bottom.

Click OK to close the dialog. Click OK in the Define Structural Interface Properties dialog.

Grouted tiebacks

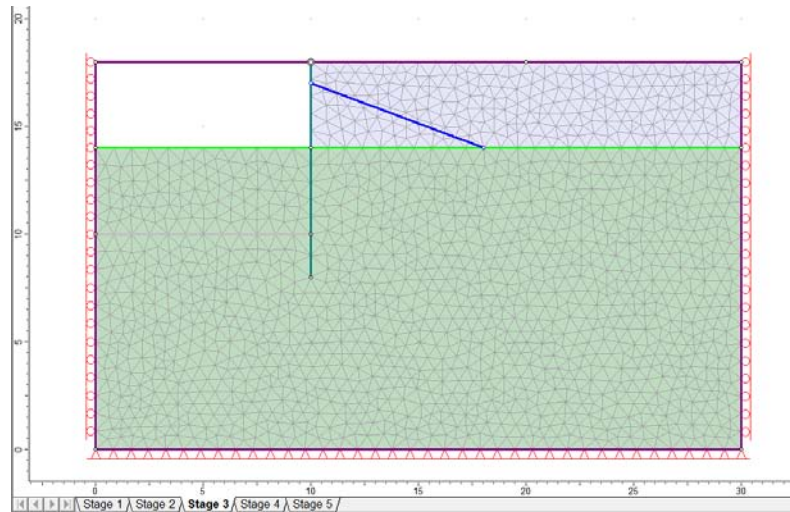


The first tieback will be added in Stage 3 simultaneous with the excavation so click on the tab for Stage 3. Go to the **Support** menu and select **Add Bolt**. You will see the Add Bolt dialog. Ensure the Bolt Property is Bolt 1 and 'Install at Stage:' is 3.



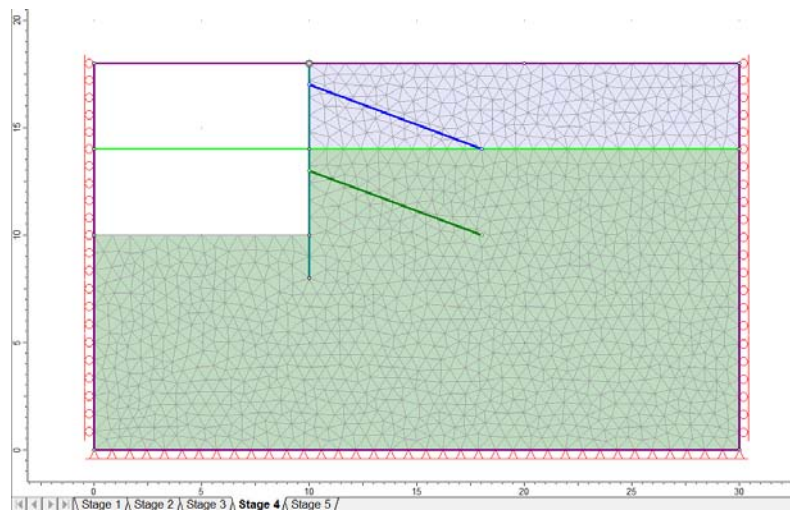
The 'Add Bolt' dialog box is shown. It has a 'Bolt Property' dropdown menu set to 'Bolt 1'. Below it is a 'Staging' section with 'Install at stage:' set to 3 and 'Remove at stage:' set to 5. There are 'OK' and 'Cancel' buttons at the bottom.

Click OK to start entering bolt coordinates. Enter 10 , 17 for the first coordinate. Enter the coordinates 18 , 14 for the second point. Hit Enter to stop entering points. The model will look like this:



Click on Stage 4 to add the next tieback. Select **Add Bolt** again from the **Support** menu and choose Bolt 2 for the Bolt Property.

Click OK to close the dialog and then enter the points 10 , 13 and 18 , 10. Hit Enter. The model should now look like this:





To set the tieback properties, select **Define Bolts** from the **Properties** menu. For Bolt 1, change the bolt type to Tieback. Change the Pre-Tensioning force to 20 kN and the Percent of Length to 40%.

The Bond Shear Stiffness and Bond Strength defaults are for a bolt anchored in bedrock. Since this bolt is anchored in sand, reduce each by a factor of 10 as shown.

Also, set the borehole diameter to 50 mm. This ensures that the ungrouted part of the bolt does not resist joint movement.

Define Bolt Properties

Bolt 1

Name: Bolt 1 Bolt Color: [Blue]

Bolt Properties

Bolt Type: Tieback

Bolt Diameter (mm): 19

Bolt Modulus, E (kPa): 200000000

Tensile Capacity (kN): 100

Residual Tensile Capacity (kN): 0

Out-of-plane Spacing (m): 1

Bond Shear Stiffness (kN/m/m): 10000

Bond Strength (kN/m): 5000

Borehole Diameter (mm): 50

Bolt Model

☐ Elastic ☒ Plastic ☒ Joint Shear

Pre-Tensioning

Pre-Tensioning Force (kN): 20 ☒ Constant Force in Install Stage

Face Plates

☒ Attached Face Plates

☐ Add Pull-Out Force

Force (kN): 0

☐ Constant Shear Stiffness

Stiffness (kN/m/m): 0

☐ Add Bulges

Define Bulges ...

Bond Length

☒ Percent of Length: 40

☐ Length (m): 1

☐ Secondary Bond Length Define...

Copy To... ☐ Show only properties used in model OK Cancel

Click on the tab for Bolt 2. This bolt material is exactly the same as bolt 1 except that the Pre-tension force is 50 kN. So change the Bolt Type to tieback, the Pre-Tensioning force to 50 kN, the Percent of Length to 40%, the Bond Shear stiffness to 10000 KN/m/m, the Bond Strength to 5000 kN/m and the borehole diameter to 50 mm.

Define Bolt Properties

Bolt 2

Name: Bolt 2 Bolt Color: █

Bolt Properties

Bolt Type: Tieback

Bolt Diameter (mm): 19

Bolt Modulus, E (kPa): 200000000

Tensile Capacity (kN): 100

Residual Tensile Capacity (kN): 0

Out-of-plane Spacing (m): 1

Bond Shear Stiffness (kN/m/m): 10000

Bond Strength (kN/m): 5000

Borehole Diameter (mm): 50

Bolt Model

☐ Elastic ☒ Plastic ☒ Joint Shear

Pre-Tensioning

Pre-Tensioning Force (kN): 50 ☒ Constant Force in Install Stage

Face Plates

☒ Attached Face Plates

☐ Add Pull-Out Force
Force (kN): 0

☐ Constant Shear Stiffness
Stiffness (kN/m/m): 0

☐ Add Bulges
Define Bulges ...

Bond Length

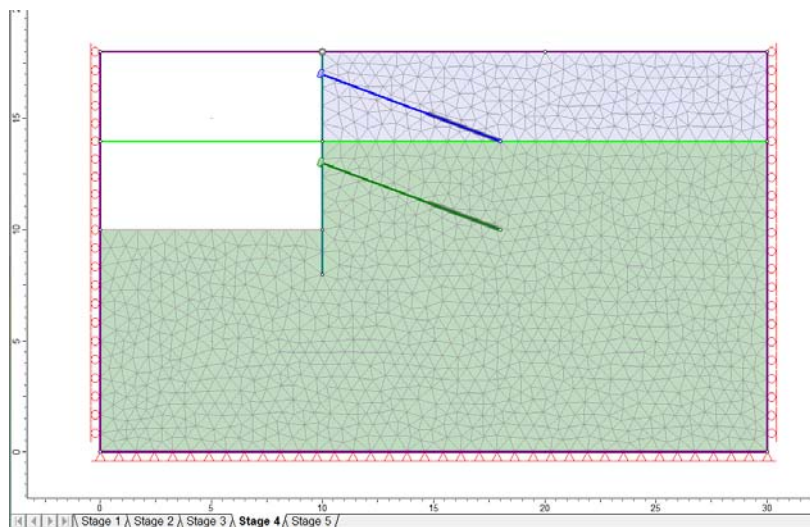
☒ Percent of Length: 40

☐ Length (m): 1

☐ Secondary Bond Length Define...

Copy To... ☐ Show only properties used in model OK Cancel

Click OK to close the dialog. You will now see that 40% of the bolts are grouted (shown as an increased line width of the bolt).



Surcharge



We wish to apply a small surcharge to the soil surface behind the wall. Click on the tab for Stage 5. Go to Loading → Distributed Loads → Add Uniform Load. Set the magnitude to 10 kN/m² as shown and click the checkbox for Stage Load.

Add Distributed Load

Orientation:

- ☒ Normal to boundary
- ☐ Vertical (+ve down)
- ☐ Horizontal (+ve right)
- ☐ Angle from horizontal
- ☐ Angle to boundary
- ☐ Field stress vector

Magnitude (kN/m²): 10

☒ Stage Load Stage Factors...

0

OK Cancel

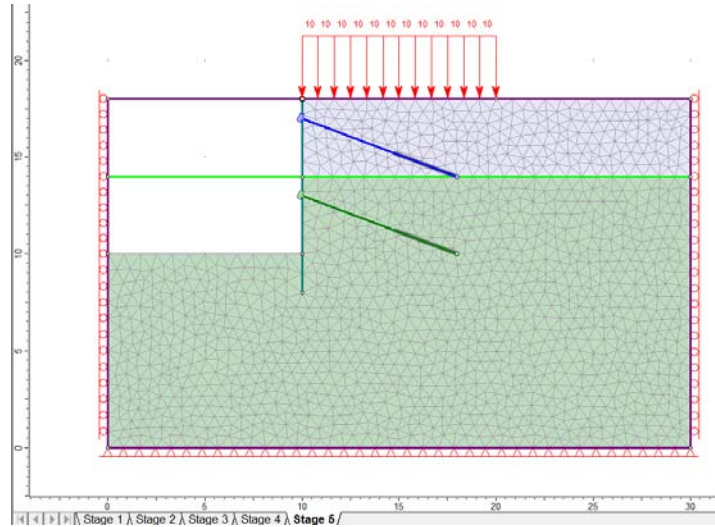
Click on the button for Stage Factors. Set all Stage Factors to 0 except for Stage 5 as shown.

Stage Factors

Stage	Factor
1	0
2	0
3	0
4	0
5	1

OK Cancel

Click OK to close the dialog. Click OK to close the Add Distributed Load dialog. You will now be prompted to select a boundary on which to apply the load. Select the top boundary just to the right of the excavation and hit enter. Your final model for Stage 5 should now look like this:



You have completed the definition of the model. Save the model using the **Save As** option in the **File** menu.

Compute



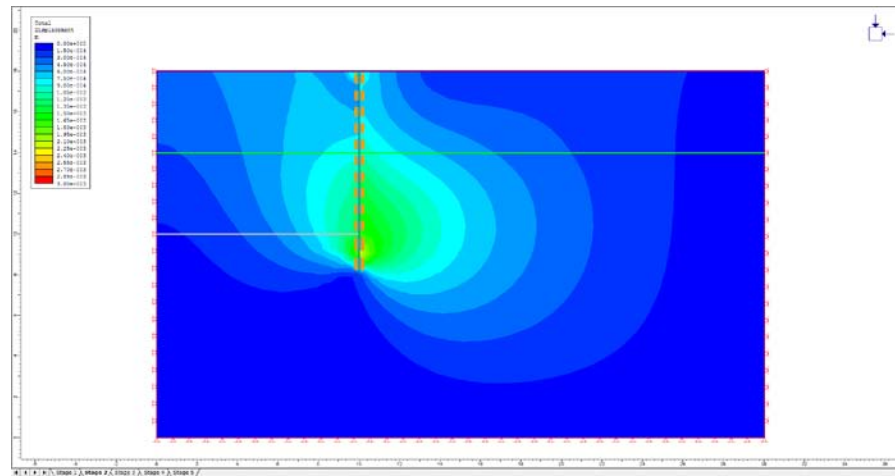
Run the model using the **Compute** option in the **Analysis** menu. The analysis should take under a few minutes to run.



Once the model has finished computing (Compute dialog closes), select the **Interpret** option in the **Analysis** menu to view the results.

Interpret

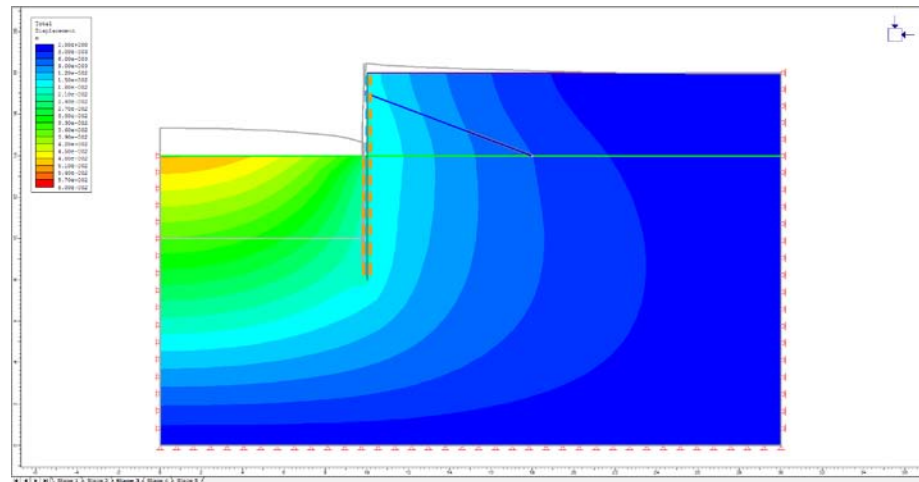
The Interpret program starts and reads the results of the analysis. You will see the maximum stress for Stage 1. Stage 1 only shows the stress due to gravity in the undisturbed material. Click the tab for Stage 2 to observe the stress after the installation of the sheet pile wall. There is not much change from Stage 1. Change the contours to plot Total Displacement (using the pull down menu at the top). The model for Stage 2 will look like this:



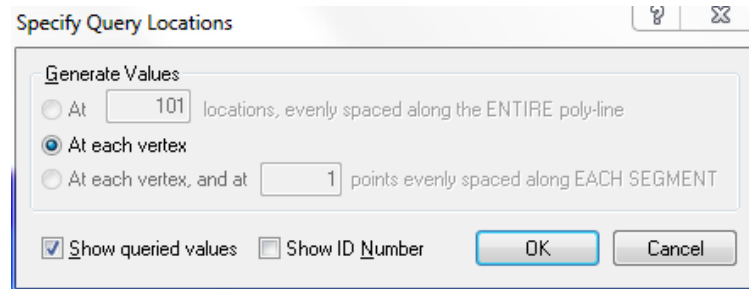
You can see that there is some displacement due to the installation of the sheet pile wall.



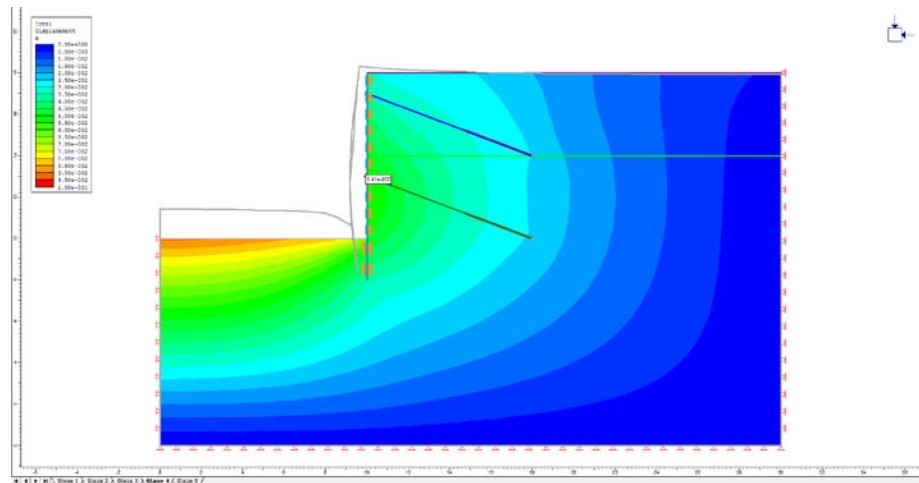
Click on the tab for Stage 3 and display the deformed boundaries by clicking on the Display Deformed Boundaries button on the toolbar at the top. Now you can see a slight bulge in the sheet pile wall and the heave of the bottom of the excavation, due to removal of the excavated material.



Click the tab for Stage 4 to show the results of the second level of excavation. Here the bulging of the wall is more pronounced. To see the actual displacement of the wall we can add a query point. Go to Query → Add Material Query and select the point at the head of the lower bolt (10,13) and hit enter. Specify the query locations as shown in the dialog below and click OK.

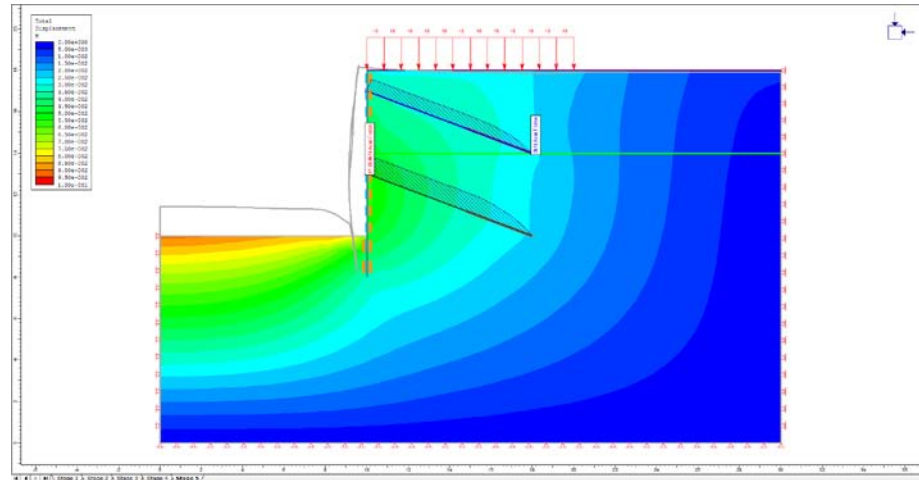


You will now see the total displacement at the point (in metres).



You will see a displacement of ~5.4 cm. Click on the tab for Stage 5 and you will see a displacement of ~5.4 cm. You may decide that this is too large for your specifications, in which case you may want to try applying a larger pretension force to the bolts.

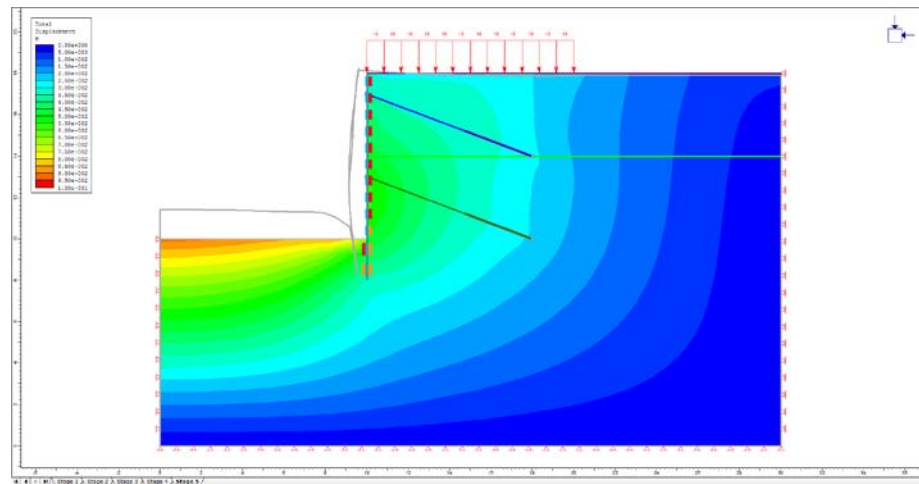
Delete the query by right clicking on it and choosing Delete Query. Right click on one of the bolts and select Show Values → Axial Force. You can now see the axial force distribution throughout the bolts.



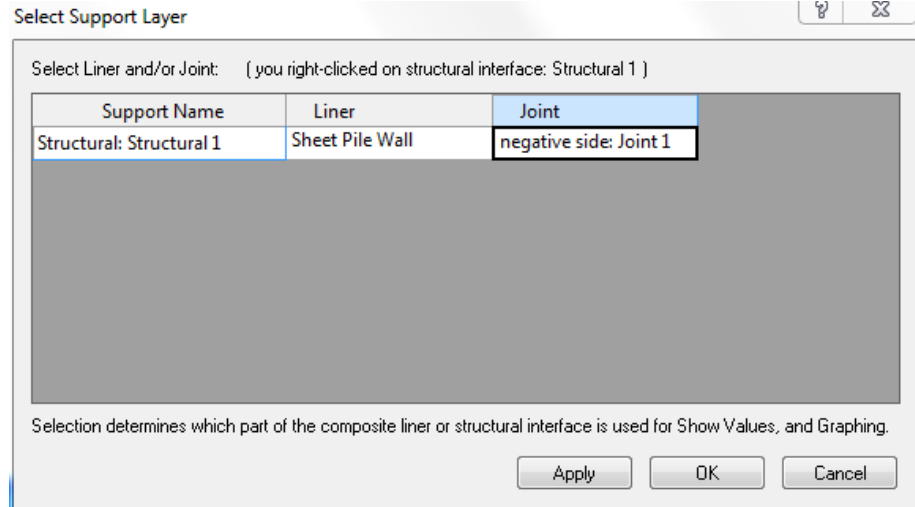
The maximum axial force in the lower bolt (~ 61 kN) is below the tensile strength (100 kN) so you could perhaps apply a slightly larger pretension without causing bolt failure.



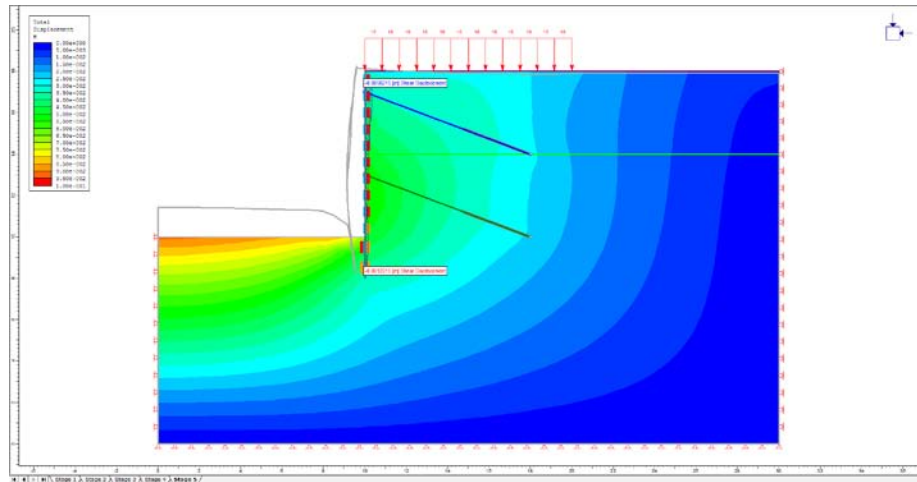
Right click on the bolts and select Show Values $\rightarrow$ Show Values (all bolts off). Now click the Display Yielded Liners button on the toolbar. The wall itself was set to be elastic so it won't exhibit any failure. However the joint between the liner and soil shows some failure (red elements), indicating that the joint is slipping or separating. It is interesting that there is little slip below the lower bolt, suggesting that the bolt is successfully preventing slip/separation from occurring.



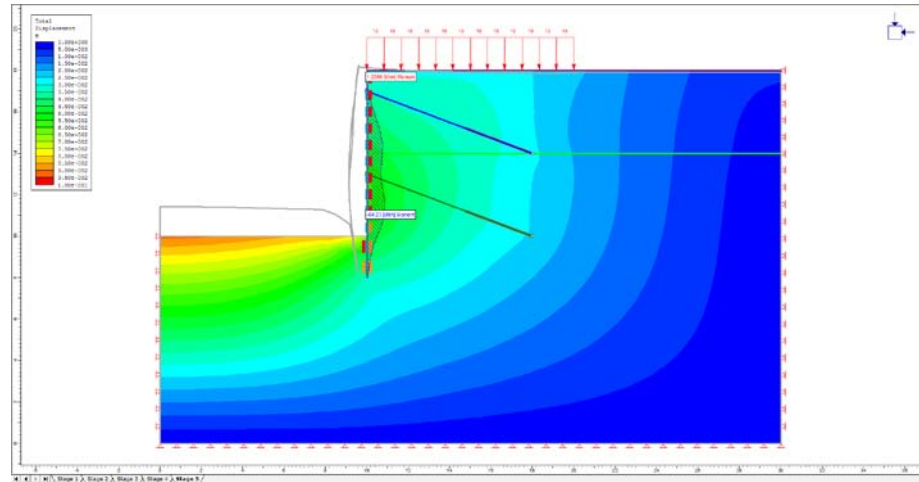
We can look at the joint slippage by following these steps. Right click on the wall and select Show Values $\rightarrow$ Joint Shear Displacement. You will see values only at the very bottom of the wall. This suggests that we are seeing the slip on the left side of the wall rather than on the right. To remedy this, right click on the wall and click Select Support Layer. Click under the Joint heading and select 'negative side: Joint 1'.



Click OK. You will now see the shear displacement between the joint and the soil to the right of the wall.



Finally, we can look at the moments in the sheet pile wall. Right click on the wall and select Show Values → Bending Moment. You can see a maximum moment of ~64 kNm.



Notice the inflection in the moment curve at the location of the lower bolt.

This concludes the tutorial, you may now exit the *Phase2* Interpret and *Phase2* Model programs.